

Genova Mongalo

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SUMMARY

AI/ML engineer and published IEEE author (M.S. in CS, AI emphasis) who builds training and evaluation datasets, defines ground-truth labels and accuracy metrics, and benchmarks deep models in PyTorch. Software-engineering and data-science experience since 2021 spanning federal AI contracting, NSF-funded research, and a production LLM product. Writes precise, publication-grade technical documentation.

EDUCATION

Georgia Institute of Technology

Atlanta, GA

M.S. in Computer Science — Artificial Intelligence emphasis Aug 2025

University of Missouri-Kansas City (UMKC)

Kansas City, MO

B.S. in Computer Science, summa cum laude Aug 2021 – Dec 2024

Relevant coursework: Deep Learning, Machine Learning, Reinforcement Learning, Artificial Intelligence, Computer Vision (CNNs/Transformers), Algorithms & Complexity, Data Structures, Brain & Cognitive Science.

EXPERIENCE

Founding AI/ML Product Engineer

Sep 2025 – Present

Geometry Health and Wellness Kansas City, MO

- Architected a full-stack stroke-support platform (**Swift/SwiftUI + Python/Flask**) delivering real-time AI-powered patient consultations, biomarker tracking, task management, and a clinician dashboard.
- Built an **end-to-end ETL pipeline** for real-time health-data processing and streaming responses, sustaining sub-second latency for urgent patient queries.
- Developed a context-aware **LLM assistant with PubMed integration** for evidence-based guidance (structured summaries, recommendations, citations); <1s WebSocket streaming inference with Ollama.
- Led product from MVP to production** — sprint cycles, technical-architecture decisions, and team coordination for HIPAA-compliant healthcare delivery.

Federal AI/ML Engineer Contractor

May 2025 – Aug 2025

AFRL Sensors Directorate Internship Program, University of Dayton Dayton, OH

- Built generative-AI (GAN) data-collection pipelines in **PyTorch** to synthesize training imagery across novel imaging modalities, improving a downstream **CNN image classifier by 50 percentage points** via a transfer-learning framework — validating synthetic-data quality against real-image ground truth.
- Implemented **end-to-end ML workflows**: data preparation, model training, evaluation, and monitoring of downstream classifier accuracy against baselines.
- Designed a **game-theoretic approach to GAN training** that measurably improved generator stability and output quality.
- Ran large-batch, parallelized training on HPC infrastructure using **Slurm**; documented code, methods, and results for cross-team review.

AI/ML Engineer Intern

Jun 2024 – Dec 2024

NSF REU, AI-Empowered Cybersecurity — University of Missouri-Kansas City Kansas City, MO

- Engineered a **Large Language Model** to detect ransomware in Industrial Control Systems, reaching **99% binary / 85% family** classification accuracy.
- Mined and queried internal datasets to construct features and ground-truth training/evaluation splits; analyzed error cases to guide model improvements.
- Sole author (with mentor guidance); work accepted and presented at **IEEE Big Data 2024** — publication-grade research writing with benchmarks and reproducible results.

Research Intern

May 2023 – Jul 2023

NSF REU, Consumer Networking — University of Missouri, Columbia Columbia, MO

- Built a **transformer** model predicting cybersickness severity on a 1–10 scale with 85% accuracy, including the data pipeline for training and evaluation.

Research Intern

Dec 2021 – Apr 2022

NASA Missouri Space Grant Consortium Kansas City, MO

- Engineered an AI-driven simulation of space transmissions to analyze signal propagation and recommend optimal frequency bands.

PROJECTS

- AWS DeepRacer — Reinforcement Learning.** Trained and tuned RL policies for a 1/18-scale autonomous race car on AWS — iteratively designed reward functions (objective, machine-gradable evaluation criteria) and measured policy performance against them.
- Phonebook Management System (Java, SQL/Postgres).** Desktop app with a relational backend: secure auth, CRUD, search, and CSV export — demonstrates database queries and data-pipeline design.

TECHNICAL SKILLS

Languages: Python, C/C++, Java/C#, SQL (Postgres), JavaScript, HTML/CSS, LISP **Spoken:** English, Spanish, Amharic

ML / CV: PyTorch, scikit-learn, pandas, NumPy, Generative AI/GANs, CNNs, Transformers, LLMs, transfer learning, reinforcement learning, benchmark & evaluation design, ground-truth dataset construction, accuracy metrics & performance monitoring

Data & Infrastructure: ETL & real-time data pipelines, SQL querying, feature engineering, data-quality monitoring, Slurm/HPC, AWS (S3/EC2/SageMaker), GCP

Tools & Frameworks: Git, Jupyter, FastAPI, Node.js, React, Swift/SwiftUI, LaTeX & Markdown technical writing, VS Code, Anaconda